

Induction Hypothesis Generalisation

Lecture Notes

Course: Computer-Assisted Theorem Proving (7CCSACTP)

1. Mathematical Induction Recipe & Generalisation Strategy

Standard Mathematical Induction Schema:

- To prove $P(n)$ holds for all $n :: \text{nat}$:
- **Base Case:** $P(0)$
- **Step Case:** $P(n) \implies P(\text{Suc } n)$ (where $P(n)$ is I.H.)

Three-Step Mechanical Strategy:

1. **Identify** property P to prove.
2. **Apply schema** (determine base and step cases).
3. **Attempt proof:**
 - If I.H. succeeds $\implies \checkmark$
 - If I.H. fails $\implies$ Generalise P to a stronger property P' (typically by adding universal quantifiers $\forall$). Return to step 2 with P' .

2. Worked Example: Proving $a \leq b \implies c \leq d \implies a \cdot c \leq b \cdot d$

Property Formulation:

- **Original property (P):** $a \leq b \implies c \leq d \implies a \cdot c \leq b \cdot d$ (Induction on d)
- **Generalised property (P'):** $\forall c. a \leq b \implies c \leq d \implies a \cdot c \leq b \cdot d$ (Stronger I.H. quantifying over c)

Base Case ($d = 0$):

- **Assumptions:** 1. $a \leq b$, 2. $c \leq 0$
- **Goal:** $a \cdot c \leq b \cdot 0$
- **Proof steps:**
 1. $c = 0$ (from *lem1* & *assump 2*)
 2. $a \cdot 0 = 0$ (from *lem2*)
 3. $a \cdot c = 0$ (from *step 2*, *subst 1*)
 4. $a \cdot c \leq b \cdot 0$ (from *lem3*) $\checkmark$

Step Case ($d \rightarrow \text{Suc } d$):

- **I.H.:** $\forall c. a \leq b \implies c \leq d \implies a \cdot c \leq b \cdot d$
- **Assumptions:** 1. I.H., 2. $a \leq b$, 3. $c \leq \text{Suc } d$
- **Goal:** $a \cdot c \leq b \cdot (\text{Suc } d)$
- **Derivation:**
 1. $c - 1 \leq d$ (from *lem4* & *assump 3*)
 2. $a \cdot (c - 1) \leq b \cdot d$ (from I.H., *assump 2* & *1*)
 3. $a \cdot (\text{Suc } (c - 1)) \leq b \cdot (\text{Suc } d)$ (*lem5*, *assump 2*, *step 2*)

Case Analysis on $c \geq 1$:

- **Case 1 ($c \geq 1$):**
 1. $\text{Suc } (c - 1) = c$ (*minus-def* & *case assumpt*)
 2. $a \cdot c \leq b \cdot (\text{Suc } d)$ (from *step 3*, *subst 4*) $\checkmark$
- **Case 2 ($c \not\geq 1$, i.e. $c < 1$):**
 1. $c < 1$ (from *lem6*)
 2. $c = 0$ (from *lem7* & *step 4* $\implies$ reduces to base case) $\checkmark$

Auxiliary Lemmas Used in Proof

- **lem1:** $x \leq 0 \implies x = 0$
- **lem2:** $x \cdot 0 = 0$
- **lem3:** $0 \leq x$
- **lem4:** $x \leq \text{Suc } y \implies x - 1 \leq y$
- **lem5:** $x \leq y \implies x \cdot l \leq y \cdot m \implies x \cdot (\text{Suc } l) \leq y \cdot (\text{Suc } m)$
- **lem6:** $\neg(x \geq y) \implies x < y$
- **lem7:** $x < 1 \implies x = 0$